Pizza Sales Analysis using SQL

13 business questions answered on one year of restaurant order data



Anushree Agarwal | Data Analyst Portfolio

INTRODUCTION

About Me & The Project

About Me

- Anushree Agarwal, aspiring Data Analyst
- Skills: SQL, Excel
- Goal: turn raw data into business decisions
- GitHub: creatoranushree-debug

About the Project

- Goal: find what drives pizza sales and revenue
- Data: 2015 orders across 4 related tables
- SQL used: joins, subqueries, window functions
- 13 questions: Basic, Intermediate, Advanced

DATASET

4 Tables, 1 Year of Orders

orders

21,350 rows

order_id
order_date
order_time

orders_details

48,620 rows

order_details_id
order_id
pizza_id
quantity

pizzas

96 rows

pizza_id
pizza_type_id
size
price

pizza_types

32 rows

pizza_type_id
name
category
ingredients

Links: orders.order_id = orders_details.order_id, orders_details.pizza_id = pizzas.pizza_id, pizzas.pizza_type_id = pizza_types.pizza_type_id

Total Orders & Revenue

Q1. Total number of orders placed

```
SELECT COUNT(order_id) AS total_orders
FROM orders;
```

Q2. Total revenue from pizza sales

```
SELECT ROUND(SUM(orders_details.quantity *
  pizzas.price), 2) AS total_sales
FROM orders_details
JOIN pizzas
  ON pizzas.pizza_id = orders_details.pizza_id;
```

21,350

total orders placed

$817,860

total revenue generated

Highest-Priced Pizza

Q3. Highest-priced pizza

```
SELECT pizza_types.name, pizzas.price
FROM pizza_types
JOIN pizzas ON pizza_types.pizza_type_id =
  pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

$35.95

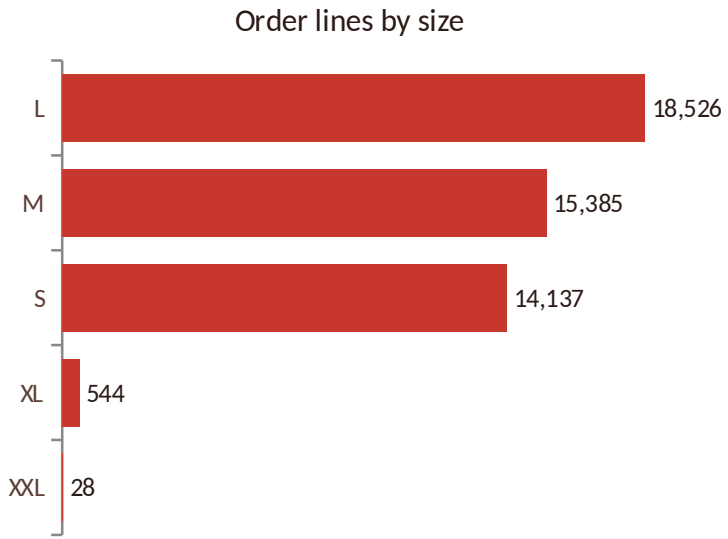
The Greek Pizza

Top price on the menu, charged for the XXL size

Most Common Pizza Size

Q4. Most common pizza size ordered

```
SELECT pizzas.size,
  COUNT(orders_details.order_details_id)
    AS orders_count
FROM pizzas
JOIN orders_details
  ON pizzas.pizza_id = orders_details.pizza_id
GROUP BY pizzas.size
ORDER BY orders_count DESC;
```

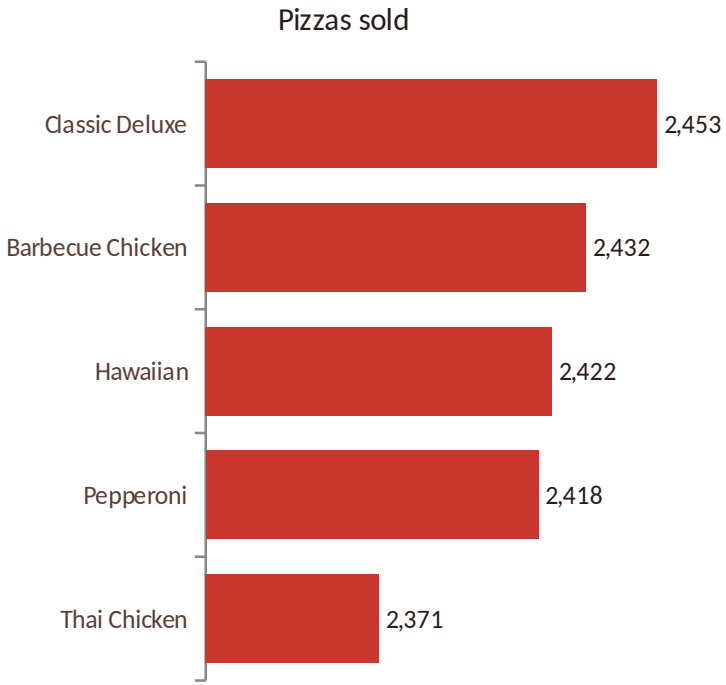


Large (L) is the most ordered size

Top 5 Most Ordered Pizzas

Q5. Top 5 pizza types by quantity

```
SELECT pizza_types.name,
  SUM(orders_details.quantity) AS quantity
FROM pizza_types
JOIN pizzas ON pizza_types.pizza_type_id =
  pizzas.pizza_type_id
JOIN orders_details
  ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5;
```



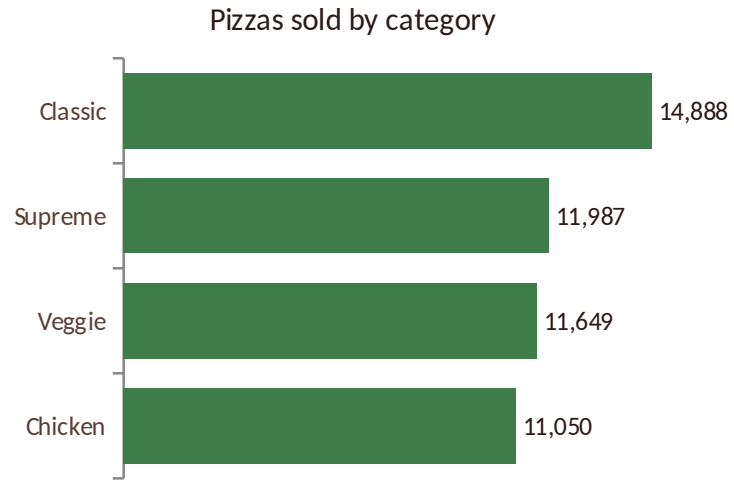
Category Quantity & Distribution

Q6. Total quantity by pizza category

```
SELECT pizza_types.category,
  SUM(orders_details.quantity) AS quantity
FROM pizza_types
JOIN pizzas ON pizza_types.pizza_type_id =
  pizzas.pizza_type_id
JOIN orders_details
  ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC;
```

Q8. Pizza types per category

```
SELECT category, COUNT(name)
FROM pizza_types
GROUP BY category;
```



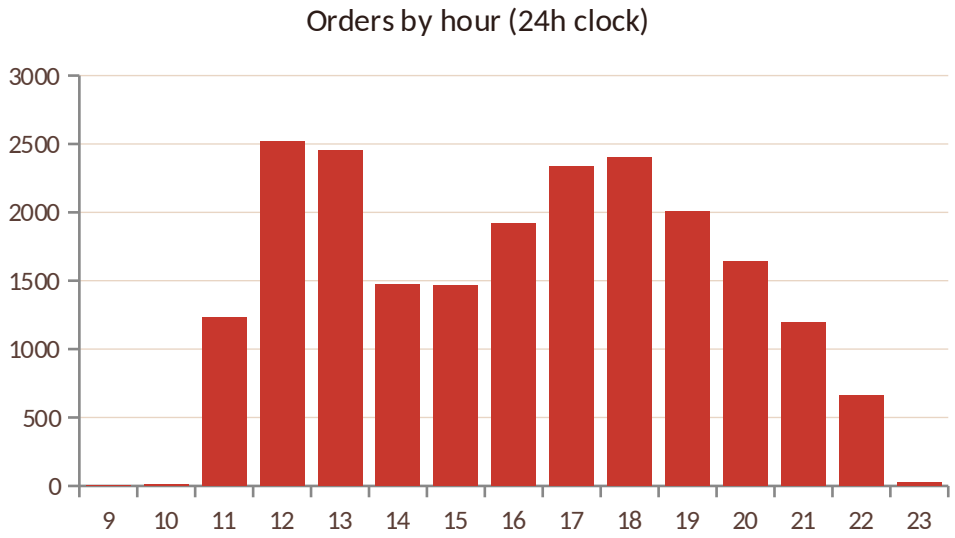
Pizza types per category

Supreme	Veggie	Classic	Chicken
9	9	8	6

Orders by Hour of the Day

Q7. Distribution of orders by hour

```
SELECT HOUR(order_time) AS HOUR,
  COUNT(order_id) AS orders_count
FROM orders
GROUP BY HOUR(order_time);
```



Two rushes: lunch (12-1 PM) and dinner (5-7 PM)

Average Pizzas Ordered per Day

Q9. Average pizzas ordered per day

```
SELECT ROUND(AVG(quantity), 0)
  AS avg_pizza_ordered_per_day
FROM (
  SELECT orders.order_date,
    SUM(orders_details.quantity) AS quantity
  FROM orders
  JOIN orders_details
    ON orders.order_id = orders_details.order_id
  GROUP BY orders.order_date
) AS order_quantity;
```

138

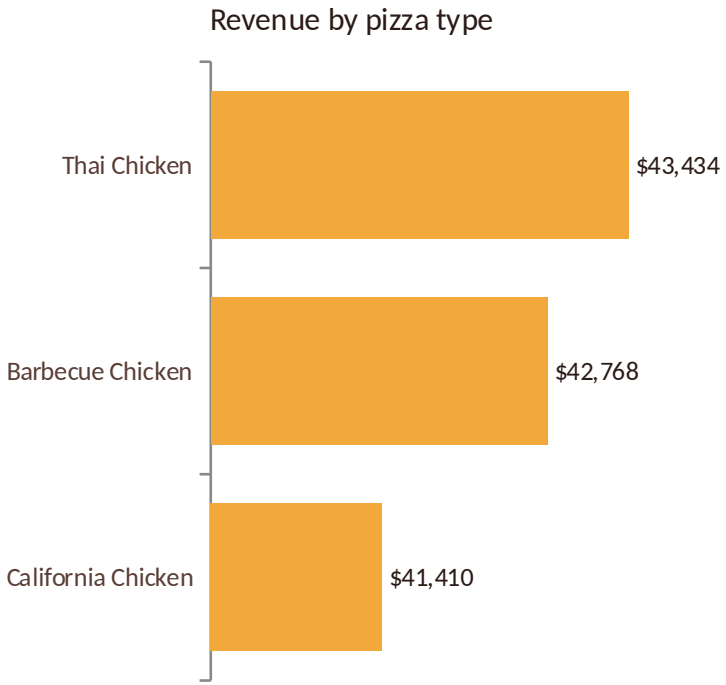
pizzas ordered per day (average)

49,574 pizzas sold across 358 order days

Top 3 Pizzas by Revenue

Q10. Top 3 pizza types by revenue

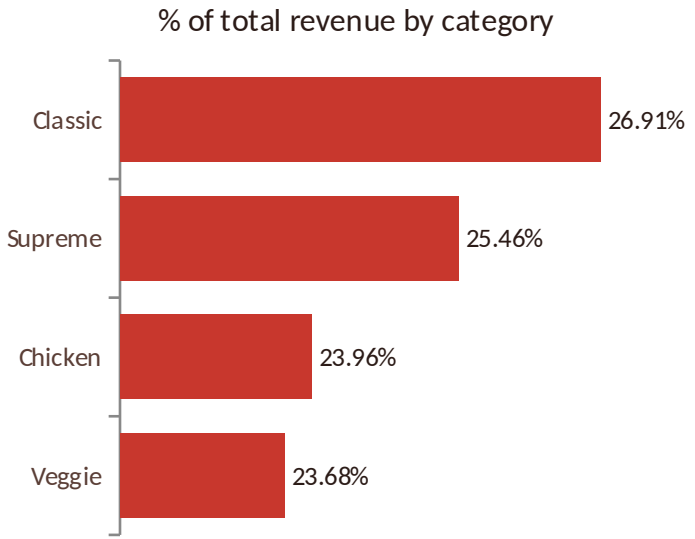
```
SELECT pizza_types.name,
  SUM(orders_details.quantity * pizzas.price)
    AS revenue
FROM pizza_types
JOIN pizzas
  ON pizzas.pizza_type_id = pizza_types.pizza_type_id
JOIN orders_details
  ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```



Revenue Contribution by Category

Q11. % contribution to total revenue

```
SELECT pizza_types.category,
  ROUND(SUM(orders_details.quantity *
    pizzas.price) / (SELECT
      ROUND(SUM(orders_details.quantity *
        pizzas.price), 2) AS total_sales
    FROM orders_details
    JOIN pizzas ON pizzas.pizza_id =
      orders_details.pizza_id) * 100, 2)
    AS revenue
FROM pizza_types
JOIN pizzas ON pizza_types.pizza_type_id =
  pizzas.pizza_type_id
JOIN orders_details
  ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue DESC;
```

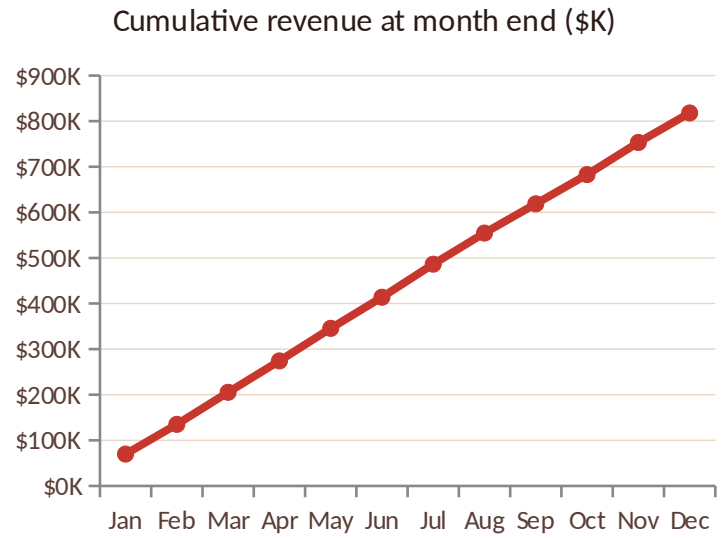


Classic leads with 26.91%; all four are close

Cumulative Revenue Over Time

Q12. Cumulative revenue by date

```
SELECT order_date,
  SUM(revenue) OVER (ORDER BY order_date)
    AS cum_revenue
FROM (
  SELECT orders.order_date,
    SUM(orders_details.quantity * pizzas.price)
      AS revenue
  FROM orders_details
  JOIN pizzas
    ON orders_details.pizza_id = pizzas.pizza_id
  JOIN orders
    ON orders.order_id = orders_details.order_id
  GROUP BY orders.order_date
) AS sales;
```



Steady growth to $817,860 by 31 Dec

Top 3 Pizzas in Each Category

Q13. Top 3 by revenue per category

```
SELECT name, revenue FROM (
  SELECT category, name, revenue,
    RANK() OVER (PARTITION BY category
      ORDER BY revenue DESC) AS rn
  FROM (
    SELECT pizza_types.category, pizza_types.name,
      SUM(orders_details.quantity * pizzas.price)
        AS revenue
    FROM pizza_types
    JOIN pizzas ON pizza_types.pizza_type_id =
      pizzas.pizza_type_id
    JOIN orders_details
      ON orders_details.pizza_id = pizzas.pizza_id
    GROUP BY pizza_types.category, pizza_types.name
  ) AS a
) AS b
WHERE rn <= 3;
```

Chicken
1. **Thai Chicken** $43.4K
2. **Barbecue Chicken** $42.8K
3. **California Chicken** $41.4K

Classic
1. **Classic Deluxe** $38.2K
2. **Hawaiian** $32.3K
3. **Pepperoni** $30.2K

Supreme
1. **Spicy Italian** $34.8K
2. **Italian Supreme** $33.5K
3. **Sicilian** $30.9K

Veggie
1. **Four Cheese** $32.3K
2. **Mexicana** $26.8K
3. **Five Cheese** $26.1K

CONCLUSION

Key Insights

Revenue & volume

$817.9K from 21,350 orders and 49,574 pizzas, about 138 pizzas a day.

Two daily rushes

Orders peak at 12-1 PM and 5-7 PM. 12 PM alone has 2,520 orders.

Size & category

Large is the most ordered size. Classic leads with 14,888 pizzas sold.

Balanced menu

Thai Chicken tops revenue (5.31%) while Classic Deluxe sells the most units.

Thank you! Anushree Agarwal | github.com/creatoranushree-debug